Table 7.3 JOSE public key encryption algorithms

JWE Algorithm	Details	Comments
`RSA1_5`	RSA with PKCS#1 v1.5 padding	**This mode is insecure** and should not be used.
`RSA-OAEP`	RSA with OAEP padding using SHA-1	OAEP is secure but RSA decryption is slow, and encryption produces large JWTs.
`RSA-OAEP-256`	RSA with OAEP padding using SHA-256	
`ECDH-ES`	Elliptic Curve Integrated Encryption Scheme (ECIES)	A secure encryption algorithm but the `epk` header it adds can be bulky. Best when used with the X25519 or X448 curves.
`ECDH-ES+A128KW`	ECDH-ES with an extra AES key-wrapping step	
`ECDH-ES+A192KW`		
`ECDH-ES+A256KW`		

WARNING Most of the JWE algorithms are secure, apart from `RSA1_5` which uses the older PKCS#1 version 1.5 padding algorithm. There are known attacks against this algorithm, so you should not use it. This padding mode was replaced by Optimal Asymmetric Encryption Padding (OAEP) that was standardized in version 2 of PKCS#1. OAEP uses a hash function internally, so there are two variants included in JWE: one using SHA-1, and one using SHA-256. Because SHA-1 is no longer considered secure, you should prefer the SHA-256 variant, although there are no known attacks against it when used with OAEP. However, even OAEP has some downsides as it’s a complicated algorithm and less widely implemented. RSA encryption also produces larger ciphertext than other modes and the decryption operation is very slow, which is a problem for an access token that may need to be decrypted many times.